

PHYS 230 - Electricity and Magnetism

Lecture 4

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

1 Electric Charge and Electric Field

- Electric Field (Continued)
- Charge Distributions

All figures in these slides are from one of the following sources, unless cited otherwise:

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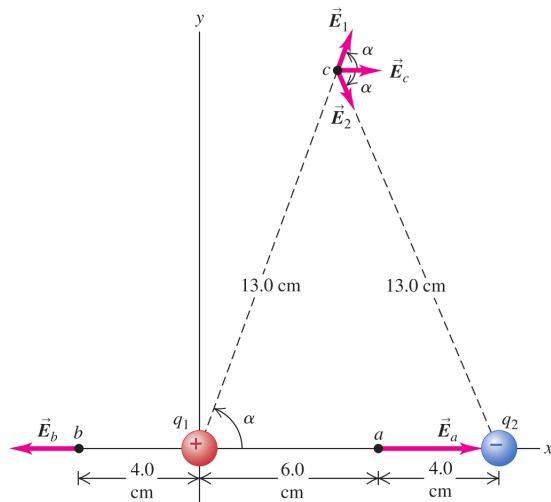
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Exercise I

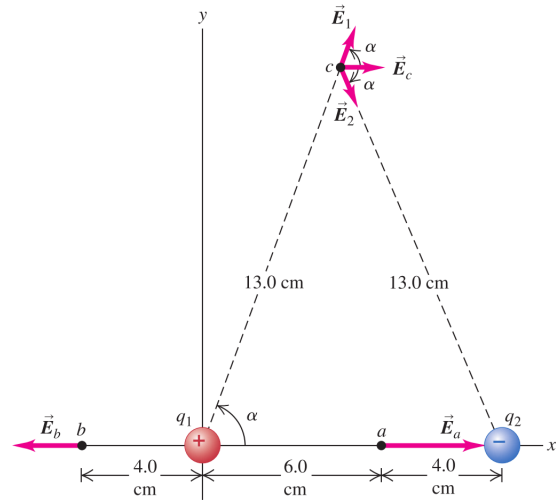
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Point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ are 0.100 m apart. Compute the electric field caused by q_1 , the field caused by q_2 , and the total field **(a)** at point a; **(b)** at point b; **(c)** at point c.



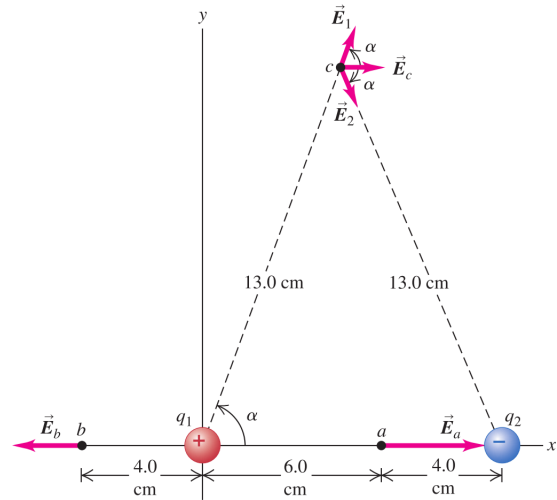
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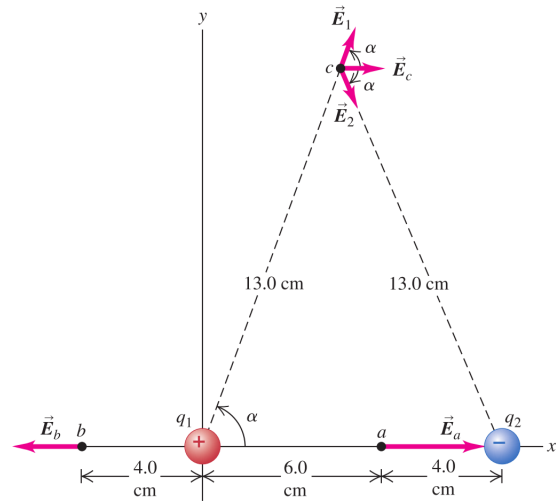
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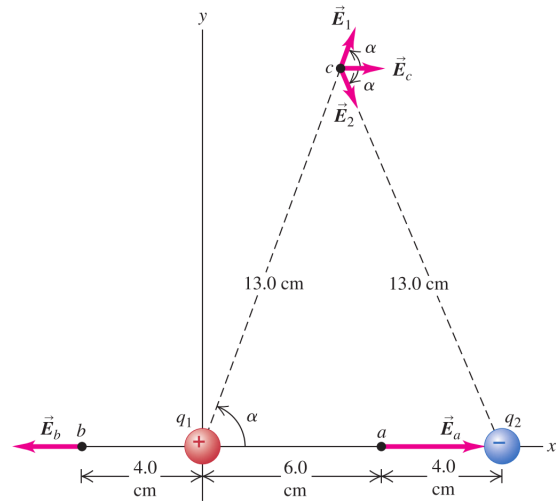
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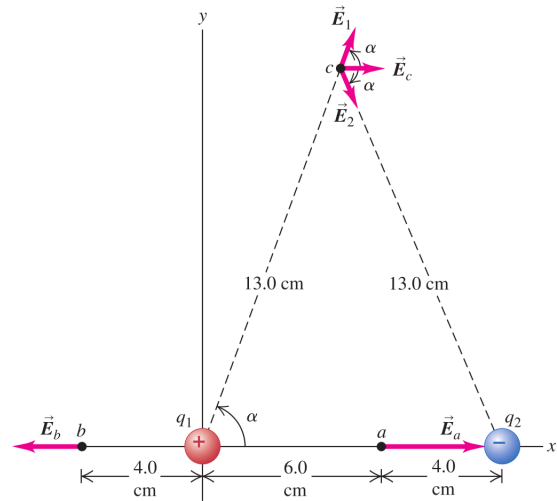
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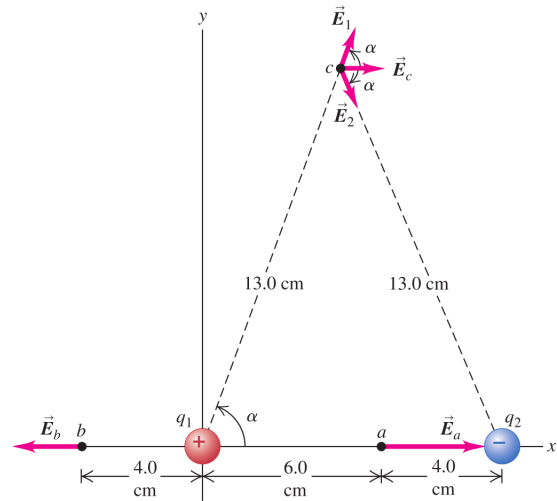
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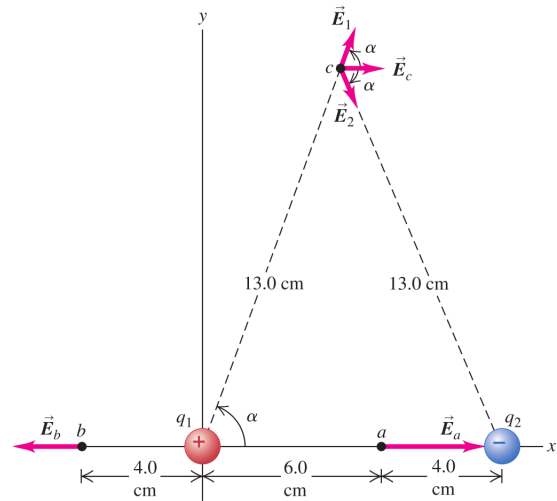
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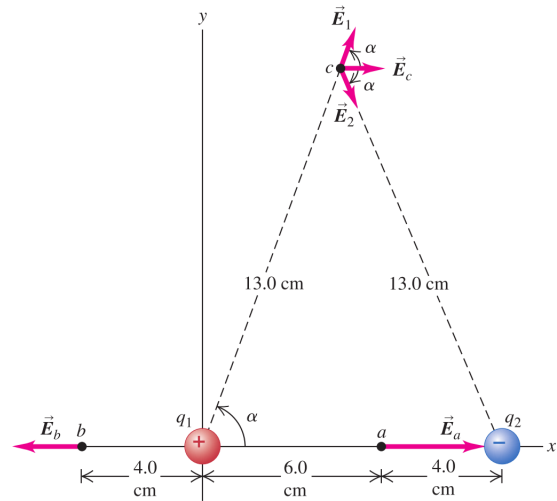
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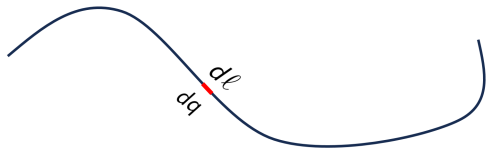
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- **Volume** charge **distribution**: charges are continuously distributed over a 3-dimensional volume

Description of Charge Distributions

Linear charge **distribution**: described by a **linear charge density**

$$\lambda = \frac{dq}{d\ell}$$

- $d\ell$ is an **infinitesimal** element of **length** on the linear distribution
- dq is an **infinitesimal** element of **charge** on the linear distribution with length $d\ell$, considered a point charge
- The units of λ is $[\lambda] = \frac{[dq]}{[d\ell]} = \frac{\text{C}}{\text{m}}$

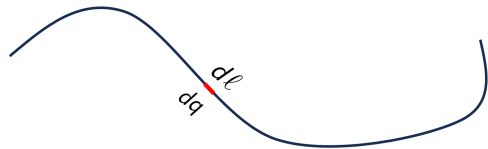


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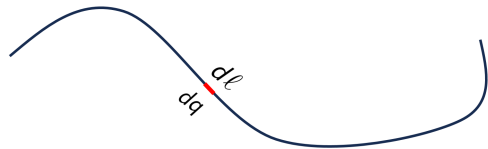


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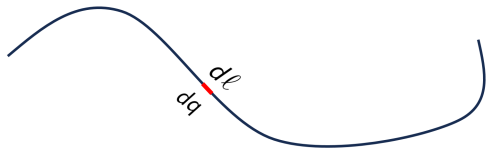
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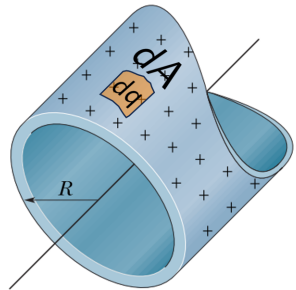
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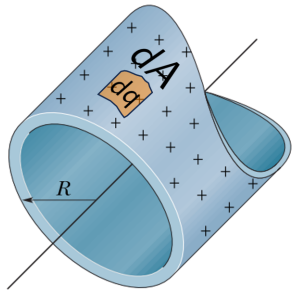


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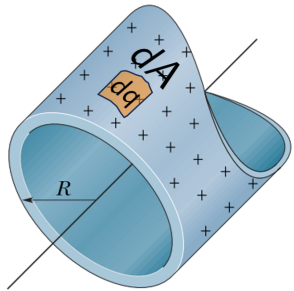


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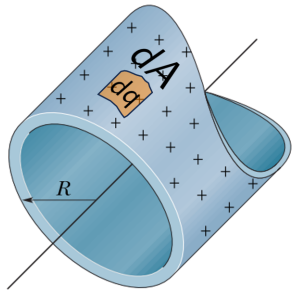


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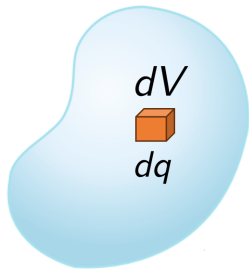


Description of Volume Distributions

Volume charge **distribution**: described by a **volume charge density**

$$\rho = \frac{dq}{dV}$$

- dV is an **infinitesimal** element of **volume** of the volume distribution
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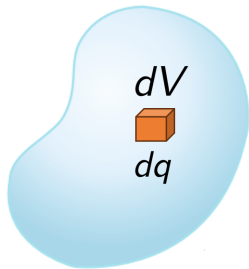


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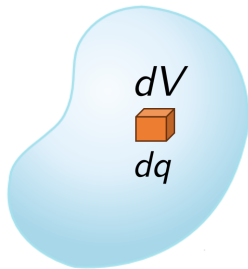


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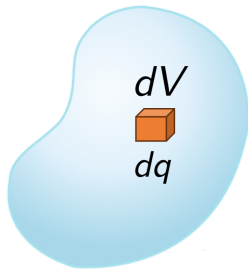


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Description of Distributions

Note: a **uniform density or distribution** of charges means

- the average density is the same as the local density, e.g., $\rho = \bar{\rho}$, which means
- the **density is constant**, e.g.

$$\rho = \text{constant}$$

Strategy To Find Electric Field/Force Vector/Magnitude in Distributions

1. Choose a coordinate system wisely
2. Choose line/surface/volume charge elements wisely
3. Depending on the problem, choose the formula (forms 1, 2, 3)
4. Write the field/force of the infinitesimal element according to step 3, for example

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \hat{\mathbf{r}}$$

5. Re-write the above formula using the charge density formula, e.g., $dq = \rho dV$ so that you get

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\rho dV}{r^2} \hat{\mathbf{r}}$$

6. Write step 5 explicitly, divide computations into simpler steps, e.g., first find $\mathbf{r}_2 - \mathbf{r}_1$, then $|\mathbf{r}_2 - \mathbf{r}_1|$, etc.
7. Integrate the result of step 6, with appropriate integration limits, to get

$$\mathbf{E} = \int d\mathbf{E} = \int \frac{1}{4\pi\epsilon_0} \frac{\rho}{r^2} \hat{\mathbf{r}} dV$$